



Prepared with Synoptic

Investigation source pack

Haul truck and light vehicle collision, Bindarra Gold Mine

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Safety reports

Two reports, quoted exactly as they were submitted: a hazard report describing this occurrence seven days before it happened, with the entry that closed it, and the haul truck operator's own account written the afternoon afterwards.

Fictional source material for the Bindarra case study. The reports, the reference numbers and the people are invented. Both are quoted exactly as they were submitted.

Hazard report HAZ-2026-0917

Submitted by: Dewatering technician, Orrabinda Resources

Submitted: 04/08/2026 23:48 local, the same night as the near miss it describes and seven days before the occurrence

Category: Hazard - traffic and mobile equipment

Risk rating applied by the submitter: Medium

Risk rating on closure: Low

Status: Closed 06/08/2026

Text as submitted:

Coming out of the new dewatering road onto the main ramp tonight about ten past ten I have come out and there was a truck coming down the ramp maybe sixty metres up. He has not seen me and I have not seen him until I was already out on the road. I have reversed back into the road and let him go past. Nothing happened but it was close enough that I am writing it up. The problem is you cannot see up the ramp from that road. There is a rise in the ramp and a windrow along the side and you are out on the road before you can see anything. There is no sign on the ramp telling the trucks the road is there and there is no give way on our side. The closed sign that was there while they were building it has been taken away. The trucks are doing forty coming down empty. I do not think they know the road is there.

Closure entry, made 06/08/2026 by the Dewatering Superintendent:

"Spoke with the dewatering crew at pre-start 05/08 and reminded them to take care entering the main ramp and to give way to haul trucks at all times. Crew acknowledged. Trucks have right of way on the ramp per the traffic management plan. No further action required. Closed."

The report was not referred to the mining department, the contractor, the survey department or the traffic management plan owner. The closure entry is the only action recorded against it.

Safety report ASR-2026-0341

Submitted by: Haul truck operator, HT-407, Tarrowen Mining Services

Submitted: 12/08/2026 16:20 local, the afternoon after the occurrence

Scheme: Submitted under the site's just culture reporting policy. The report may inform the investigation; it is not to be quoted into a distributed document.

Text as submitted:

Night ten of fourteen, second cycle after crib, coming down the main ramp empty in 407 at about thirty-seven. I came over the crest above the three-twelve and there was a light vehicle out on the ramp in front of me. By the time I saw the whip he was about forty metres away. I got on the brakes and steered right into the windrow, which was all I had, and I hit him on the left front. He went down the ramp in front of me and rolled over against the inside windrow. I called it in and stayed in the cab until dispatch stopped the ramp, then went down to him and stayed with him until the ERT got there.

I want to put three things down while they are clear.

I did not know that road was open. I knew they had cut it. There was a closed sign on it and then there was not, and nobody told me anything about it either way. It is not on my in-cab screen. As far as I knew I was driving down a ramp with no intersection on it.

I did not make a radio call before that spur and nor does anybody else. The list of intersections you call at is on the wall in the crib room and that one is not on it. I am not offering that as an excuse. I am saying that if you asked every operator on this pit what the rule is, they would tell you the same thing I am telling you, and the rule is doing exactly what it says it does.

The last thing is the collision awareness. I have been on this pit fourteen months. That system goes off two or three times a shift up on the haul road and it has never once gone off down in the pit. I took that to mean the pit was empty when it was quiet. I found out the day after that it is switched off down there on purpose. Nobody ever told me. I would like to know how many other operators think what I thought, because I would guess all of them.

I have gone over the forty metres many times since. I do not think there was anything else I could have done with it once I had it. What I keep coming

back to is that I should have had more than forty metres.

Occurrence notification

The notification raised by the night shift superintendent 39 minutes after the collision: what happened, the injuries, the damage, and the actions taken straight away.

Fictional. Orrabinda Resources, Tarrowen Mining Services, Hallorn Survey & Geotechnical, the Bindarra Gold Mine, the people, the fleet numbers and every document reference in this file are invented for teaching purposes. Any resemblance to a real operation or occurrence is coincidental.

Notification reference: ORR-OCC-2026-0332

Raised by: Night Shift Superintendent, Bindarra Gold Mine

Raised at: 12/08/2026 03:26 local, UTC+8 (11/08/2026 19:26 UTC)

Initial risk rating: High

Investigation required: Yes (risk rating Medium or above)

Statutory notification: Reportable injury, notified to the State Mines

Inspectorate at 12/08/2026 04:05 local (11/08/2026 20:05 UTC).

The occurrence

Item	Detail
Date	12/08/2026 local (11/08/2026 UTC)
Time	02:47:12 local, UTC+8 (18:47:12 UTC)
Location	Bindarra Gold Mine, Stage 4 open pit, main ramp at the RL 312 dewatering spur intersection
Operation	Orrabinda Resources Ltd (mine operator and principal employer)
Mining contractor	Tarrowen Mining Services (load and haul, drill and blast)
Equipment 1	Rigid-frame haul truck, 220 t class, fleet number HT-407, empty, descending the ramp
Equipment 2	Light vehicle, 4WD utility, fleet number LV-118, one occupant
Persons involved	Haul truck operator (Tarrowen Mining Services); grade control technician (Hallorn Survey & Geotechnical), driver of LV-118
Injuries	One serious. The driver of LV-118 sustained a fractured left clavicle, three fractured left ribs and a small pneumothorax, treated at the surface medical facility and flown to the metropolitan trauma centre at 07:40 local. No other injuries.
Damage	LV-118 destroyed: right side crushed, roll-over protection structure deformed, vehicle rolled onto its left side. HT-407: front left bumper, access ladder and left front mudguard damaged; returned to service 14/08/2026.
Production impact	Stage 4 pit stood down for 9 hours 15 minutes. Main ramp closed below RL 340 for 22 hours.

Brief description

HT-407 was descending the main ramp empty, returning to the Stage 4 loading face for its second load after the crib break. LV-118 was travelling outbound on the RL 312 dewatering spur, a 240 m access road cut under works order WO-26-4471 to reach the new dewatering bore, completed on 31/07/2026 and opened to light vehicles on 02/08/2026. It was returning to

the ramp after the grade control technician had completed a pickup at the bore collar.

The spur joins the main ramp on the truck's left, 34 m below a crest.

Neither driver knew the other was there until the truck came over the crest and its operator saw the light vehicle's beacon, about 53 m above the spur.

The haul truck operator applied retard and service braking and steered right, towards the wall. The

truck's front left struck the right side of LV-118, pushed it approximately 18 m down the ramp and rolled it onto its left side against the inner windrow.

No positive radio call was made by either driver before entering the intersection. The RL 312 spur intersection does not appear in the Bindarra traffic management plan and is not one of the nine intersections at which the plan requires a positive call.

The driver of LV-118 was conscious throughout and was extricated by mine emergency response through the rear of the vehicle at 03:14 local, 27 minutes after the collision.

Immediate actions taken

- Stage 4 pit stood down. Main ramp closed below RL 340 and the RL 312 dewatering spur barricaded and closed to all traffic.
- Both vehicles isolated and left in position for the investigation. Scene photographed and surveyed before recovery.
- Haul truck operator and the shift supervisor stood down for the remainder of the shift. Both tested for drugs and alcohol; both results negative.
- Fleet management system and collision awareness system data downloaded from HT-407 by the contractor's technical services group at 09:30 local and provided to the investigation unopened.
- Site-wide direction issued by the Mining Manager at 05:10 local: a positive call on the pit channel is required before entering any intersection in the Stage 4 pit, listed in the plan or not, until further notice.

Fleet management and collision awareness system download

The contractor's technical services download of the haul truck's positional and engine data and of its collision awareness system log, which fixes the speeds, the braking and what the detection system did.

Fictional source material for the Bindarra case study. The systems, the fleet numbers, the reference numbers and the readings are invented.

Report reference: TRW-TS-2026-114

Prepared by: Technical Services, Tarrowen Mining Services

Issued: 18/08/2026

Scope: Fleet management system (FMS) positional and engine data for HT-407, and the collision awareness system (CAS) event log for the same unit, for the period 18:30 to 19:00 UTC on 11/08/2026. Data downloaded on site at 09:30 local 12/08/2026 and provided to the investigation sealed. LV-118 carries a CAS transponder only and records nothing of its own.

Key parameters

Haul truck HT-407, descending the main ramp, empty.

Time (UTC)	Speed	Position	Note
18:45:58	39 km/h	RL 384, main ramp	Second cycle after the crib break; tray down, empty
18:46:31	38 km/h	RL 351	Retarder on the second position, steady descent
18:47:06.0	37 km/h	RL 317, 53 m above the spur mouth	Last steady data point before the brake application
18:47:07.4	37 km/h	RL 316, 38.5 m above the spur mouth	Service brake pressure rises from 0 to 620 kPa in 0.4 s; retarder to maximum; steering input 14 degrees right
18:47:09.2	32 km/h	RL 314, 21.4 m above the spur mouth	CAS proximity event logged (see below); no alert raised in cab
18:47:12.0	24 km/h	RL 312, spur intersection	Deceleration spike; collision
18:47:19	0 km/h	19 m below the intersection	Vehicle stationary, park brake applied

Braking analysis: 1.4 seconds elapsed between the last steady-state data point and the first brake pressure rise, which is the operator's response time on this download. Mean deceleration from the brake application to the collision was 0.78 m/s² net of the grade (37 km/h to 24 km/h in 4.6 seconds, over 39 m of road), against 0.80 m/s² recorded for this unit in the monthly brake performance test on 03/08/2026. On a 9.1 per cent descent gravity takes about 0.88 m/s² off what the brakes produce, which is why an

empty truck's net figure on this ramp sits near 0.8. The retarder and the service brakes performed as specified throughout. Had no braking been applied the collision speed would have been 37 km/h rather than 24 km/h. The impact and the push took the truck to about 20 km/h, and it came to rest about seven seconds after the collision.

Road network: the FMS in-cab map for the Stage 4 pit was last published on 24/07/2026. It does not contain the RL 312 dewatering spur, which was cut on 31/07/2026. The in-cab display therefore showed the main ramp with no intersection at RL 312 on the night of the occurrence. Publishing a revised road network is a survey task raised by a management of change record; no such record exists for the spur.

Light vehicle LV-118. LV-118 is not dispatched and carries no FMS. Its CAS transponder is a passive tag. Its movements are known only from the CAS proximity log, the driver's account and the scene evidence.

Collision awareness system

Configuration as found on HT-407. System operating at Level 8 (operator advisory). Level 9 (automated intervention) is available on this fleet and is not enabled at Bindarra. Advisory alerts are raised on a vehicle tag entering the 60 m detection ring at a closing rate above 10 km/h, except inside a suppression geofence. Inside a suppression geofence no alert is raised at any range, and the system logs a proximity event only once the range falls below 25 m.

Suppression geofence "S4-LOAD". Drawn 14/03/2025 around the Stage 4 loading face to stop nuisance alerts between the excavator, the trucks under it and the service vehicles working in the same area, where the alert rate made the system unusable. Extended northward on 09/06/2026 when the loading face advanced, which carried the boundary up the main ramp for the first time. The extended boundary lies 180 m up-ramp of the RL 312 spur intersection, so the intersection, cut seven weeks later, sits inside the suppressed area. The configuration record gives the new boundary and not the old one, so the size of the extension is not known.

The extension was made by the contractor's technical services group as a system configuration task. It is not recorded in the traffic management plan and was not reviewed against it.

Event log, 11/08/2026.

18:47:09.200 PROXIMITY HT-407 tag LV-118 range 21.4 m closing 32 km/h

zone S4-LOAD alert SUPPRESSED (geofence)

18:47:12.000 IMPACT HT-407 longitudinal 1.9 g logged by accelerometer

18:47:12.100 PROXIMITY HT-407 tag LV-118 range 0.0 m

zone S4-LOAD alert SUPPRESSED (geofence)

The proximity event at 18:47:09.2 was logged 2.8 seconds before the collision, when LV-118 was at the spur mouth and the truck 21.4 m above it, so the closing rate is the truck's own speed. It was logged then because that is when the range fell below 25 m. Outside a geofence the system would have raised an advisory tone and a screen indication when the tag entered the 60 m ring, which on the download is at about 18:47:06.5: half a second after the beacon came into view and about a second before the brakes went on.

What the data does and does not settle

It settles the truck's speed, the braking, the 1.4 second response and the fact that the brakes worked. It settles that the CAS saw the light vehicle and said nothing. It does not record when either driver first saw the other, because neither vehicle records a line of sight; the survey puts the first point at which the beacon was visible at 53 m, which is where the truck was at the last steady data point. It does not settle the light vehicle's speed before it reached the spur mouth.

Pit channel radio extract

A verbatim transcript of every transmission on the Stage 4 pit channel from the shift's return from crib to two minutes after the collision.

Fictional source material for the Bindarra case study. The channel, the call signs and the transmissions are invented.

Reference: RAD-2026-0332, transcribed 17/08/2026, verified by a second investigator against the recording.

Source: Bindarra Stage 4 pit channel (UHF channel 12), logged continuously at the surface repeater. Times are UTC, taken from the repeater's own clock, checked against the fleet management system clock and found to be within 0.4 seconds. The site's working clock is local time, UTC+8, so each time below is eight hours behind local time.

Period transcribed: 18:20:00 to 18:52:00 UTC on 11/08/2026 (02:20:00 to 02:52:00 local on 12/08/2026).

The investigator listened to the whole 32 minutes. The extract below is every transmission in the period from the shift's return from crib to two minutes after the collision. Nothing has been left out of the period shown.

18:22:41 Dispatch All Stage 4 units, crib's done, back to work please.
18:22:49 HT-411 411 rolling.
18:22:53 HT-407 407 rolling, first load.
18:26:05 HT-407 407 loaded, heading up.
18:28:10 HT-407 407, three-sixty crossover, loaded, coming up through.
18:28:14 Dispatch Copy 407, you're clear.
18:33:02 Excavator 2 Dig's getting tight on the north side, I'll need the dozer before the next pass.
18:33:14 Dispatch Copy, I'll task 62 when he's free.
18:37:20 HT-407 407, ROM pad turn-off, turning in.
18:37:23 Dispatch Copy 407.
18:40:02 HT-407 407 dumped, coming back down.
18:41:30 HT-411 411 loaded.
18:44:02 HT-411 411, three-sixty crossover, loaded, coming up.
18:44:05 Dispatch Copy 411, clear.
18:46:12 Excavator 2 Who've I got next?
18:46:18 Dispatch 407's about four minutes off you.
18:46:21 HT-407 407, three-sixty crossover, empty, going down through.
18:46:24 Dispatch Copy 407.
18:47:41 HT-407 [raised voice] Dispatch, 407. I've hit a light vehicle on the ramp at three-twelve. I've hit a

light vehicle. He's on his side.

18:47:49 Dispatch 407, say again?

18:47:52 HT-407 I've collided with a light vehicle at the three-twelve spur. He's rolled. I need ERT and I need them now.

18:48:01 Dispatch All Stage 4 units stop. All units stop. Emergency on the pit channel. 407, are you out of your truck?

18:48:09 HT-407 I'm getting out now.

18:48:14 Dispatch Negative 407, stay in the cab until I've stopped the traffic. Give me thirty seconds.

18:48:44 Dispatch 407, ramp's clear above and below you, you're safe to get out. ERT are rolling from the surface.

18:49:02 Supervisor Supervisor's en route from the ROM pad, five minutes. 407, is he conscious?

18:49:30 HT-407 He's conscious. He's talking to me. He can't get out, the door's against the ground.

18:49:38 Supervisor Don't move him and don't try to get him out. Wait for ERT.

18:51:20 ERT 1 ERT 1 entering the pit, RL three-eighty.

What the recording shows

The haul trucks called at every Appendix B intersection they passed in the period, and each call was acknowledged: HT-407 at the RL 360 crossover going up at 18:28:10, at the ROM pad turn-off at 18:37:20 and at the crossover coming down at 18:46:21, 51 seconds before the collision, and HT-411 at the crossover at 18:44:02.

There is no transmission by either vehicle relating to the RL 312 spur, the dewatering bore, or a light vehicle being anywhere in the Stage 4 pit, at any point in the 32 minutes transcribed. LV-118 made no transmission on the pit channel at all that shift; its only recorded transmission is at 10:04 UTC (18:04 local), logging on to the channel at the start of shift.

There is no transmission asking a haul truck to hold, and none offering a light vehicle a gap. The dispatcher's traffic at 18:46:18 places HT-407 about four minutes from the loading face, which is consistent with the truck passing RL 312 at the time it did.

The dispatcher's stop-the-pit call at 18:48:01 is 49 seconds after the collision and 20 seconds after the operator's first call. The instruction to the operator to stay in the cab until the ramp was stopped is the pit channel emergency procedure and was followed.

Scene examination and survey record

The scene as examined before recovery, the survey department's pickup of the intersection and its sight distances, and the light, surface and dust records for the night.

Fictional source material for the Bindarra case study. The measurements, the reference numbers and the records are invented.

Reference: INV-2026-0332-SCENE

Scene examined: 12/08/2026, 06:20 to 11:40 local, before recovery.

Survey pickup: 13/08/2026, by the mine survey department at the investigation's request, reference SUR-26-1188.

Scene examination

The main ramp at RL 312 runs approximately north-west on a 9.1 per cent descending grade, 24 m wide between windrows, surface dry and well maintained, no corrugation or loose material beyond the normal running surface. Traffic on the ramp keeps left, so a descending truck runs on the open, pit side of the road and an ascending vehicle on the wall side. The RL 312 dewatering spur joins the ramp on a descending truck's left, the open side, on the inside of a left-hand bend, at an included angle of 71 degrees. LV-118 was turning right out of the spur to go up the ramp, which took it across the descending lane with its right side towards a truck coming down.

Marks recovered, measured from the up-ramp edge of the spur mouth:

Mark	Detail
HT-407 left front tyre, braked	Continuous scuff commencing 38.5 m up-ramp of the spur mouth and running 58 m to the truck's resting position
HT-407 steering	The scuff deviates 1.4 m to the right over its first 20 m, consistent with the 14 degree steering input on the download
Impact position	6.2 m into the ramp from the spur mouth, 3.1 m out from the toe of the inner (pit side) windrow
LV-118 gouge	Continuous gouge from the impact position running 18.4 m down-ramp, changing at 11.2 m from tyre scuff to body panel gouge, which is where the vehicle went over
LV-118 rest position	On its left side, against the inner windrow, 18.4 m down-ramp of the impact position, facing down-ramp
Debris	Right-side glass and mirror fragments from 4.8 m to 14 m down-ramp of the impact position

LV-118 damage is concentrated on the right side from the B pillar forward.

The roll-over protection structure is deformed inward at the right rear mounting but was not breached. The driver's seatbelt was found fastened and the webbing showed loading marks. The 4.5 m amber beacon whip was sheared at its base and recovered 9 m from the vehicle; its lamp was tested after

recovery and worked.

HT-407 damage is confined to the front left bumper, the left mudguard and the lower two steps of the access ladder. There is no damage above the bumper line.

Survey of the intersection

The survey department picked up the intersection, the crest and the eye positions of both vehicles.

The spur meets the ramp 34 m below a convex grade break, where the descent steepens, which the crews call the crest. An operator descending the ramp, seated at an eye height of 5.80 m above the running surface, first has line of sight to the spur mouth at 41 m. The top of a 4.5 m beacon whip on a light vehicle approaching the mouth in the spur is in view from the same eye height from 53 m, over the grade break and above the windrow; the vehicle's body is not. A driver

at the spur mouth, at an eye height of 1.45 m, first has line of sight to a descending truck at 38 m; the inner windrow, 1.6 m high, stands between the spur and the ramp for the last 25 m of the spur's approach and takes the truck's lower body out of view before that.

The Bindarra traffic management plan, clause 6.2, requires that at every intersection the available sight distance shall be not less than the stopping distance of the largest vehicle using the road at the posted speed, and in no case less than 120 m. The stopping distance of a 220 t class truck travelling empty at the posted 40 km/h on a 9 per cent descending grade, calculated with the manufacturer's data and a 1.5 second response, is 96 m. The available sight distance at this intersection is 41 m. The intersection does not meet clause 6.2 on either limb.

Signage and delineation

At the time of the occurrence the RL 312 spur intersection carried:

- no intersection warning sign on either ramp approach;
- no give-way or stop control at the spur mouth;
- no delineation, reflective marker posts or windrow gap markers;
- no posted speed for the spur itself.

A single "ROAD CLOSED - CONSTRUCTION" sign had stood at the spur mouth while the road was being cut. The works log records it removed on 02/08/2026, the day the bore was energised and the spur was first used by light vehicles. Nothing was installed in its place.

The traffic management plan, clause 6.4, requires an intersection warning sign 50 m in advance on each approach, a give-way control on the minor road, and delineation at the mouth. Clause 6.4 applies to intersections shown on the plan's own road layout drawing. The RL 312 spur is not on that drawing.

Conditions, light and dust

Light. The occurrence was at 02:47 local, in darkness. There is no fixed lighting anywhere in the Stage 4 pit; the only lighting is vehicle lighting and the light plants at the loading face, 1.1 km away and out of sight behind the crest. Moonset was at 23:18 local the previous evening.

Surface and weather. Bindarra met station, ROM pad: wind south-east 8 km/h, temperature 9 degrees, no precipitation. The last recorded rainfall at the site was 26 days before the occurrence. The investigator examined the ramp surface at 06:20 local and found it dry throughout, with no damp patches, ponding or greasy film at or above the intersection.

Water cart and dust. The water cart log records the main ramp between RL 280 and RL 400 last watered at 01:10 local (17:10 UTC), 1 hour 37 minutes before the occurrence. The standing instruction on night shift is a ramp pass every 90 minutes, so a pass was 7 minutes overdue. The fixed dust monitor on the RL 328 bench beside the ramp, 180 m up-ramp of the intersection and above the grade break, logged PM10 at 142 micrograms per cubic metre in the 02:40 to 02:50 local interval (18:40 to 18:50 UTC), against a shift mean of 88 and a site trigger of 250. That is a measure of airborne particulate at the monitor head. Nothing at Bindarra measures visibility, and no measurement exists of what could be seen from either cab on the night.

Site procedures and policy extracts

Verbatim extracts from the statutory guideline, the traffic management plan, the management of change procedure, the works order that cut the road and the contractor management standard, transcribed from the copies current on the day.

Fictional source material for the Bindarra case study. The guideline, the regulations, the plans, the procedures and the reference numbers are all invented. Nothing here reproduces any real instrument.

Transcribed from the controlled copies current on 11/08/2026, checked against the document register on 19/08/2026.

1. Mines Safety Guideline MSG-14, "Surface mobile equipment and traffic management"

Issued by the State Mines Inspectorate, revision 4, 2021. A guideline made under regulation 6.22 of the Mining Operations Safety Regulations 2018. Compliance with the guideline is one way of discharging the regulation; the regulation itself requires only that "the operator shall manage the risks arising from the interaction of vehicles at the mine".

3.2 The operator shall prepare and maintain a traffic management plan for the mine. The plan shall be reviewed at intervals not exceeding twelve months, and following any significant change to the mine's road network, vehicle fleet or operating pattern.

3.5 The plan is a site document. It is not approved by, and need not be submitted to, the Inspectorate, except on request following an occurrence or an inspection.

5.1 The operator shall nominate the intersections, crossings and other interaction zones at the mine at which positive communication between vehicles is required before entry, and shall record them in the plan.

5.4 Where a light vehicle and a heavy vehicle may interact, the operator shall provide for the light vehicle to be detectable by the heavy vehicle operator. Detectability may be provided by line of sight, elevated marking, radio communication, a proximity detection system, or a combination of these.

The guideline does not define "significant change" at 3.2, and contains no example of one. It does not say who nominates the interaction zones at 5.1 or on what basis, and sets no minimum. It does not say which of the five means at 5.4 must be provided, or how many.

2. Bindarra Traffic Management Plan BIN-TMP-004, revision 7

Owner: Mining Manager, Orrabinda Resources. Issued 12/11/2025. Next review due 12/11/2026.

4.1 Light vehicles shall give way to heavy vehicles at all times and in all circumstances.

4.3.1 A positive call shall be made on the relevant pit channel before entering any intersection listed at Appendix B, and entry shall not be made until the call is acknowledged or the channel is confirmed clear.

4.3.2 At intersections not listed at Appendix B, vehicles shall give way in accordance with clause 4.1. No call is required.

6.2 At every intersection, the available sight distance shall be not less than the stopping distance of the largest vehicle using the road at the posted speed, and in no case less than 120 m. Where this cannot be achieved, the intersection shall be controlled by a stop control, signalised control or traffic controller.

6.4 Every intersection shown on the road layout drawing at Schedule 1 shall be provided with an intersection warning sign 50 m in advance on each approach, a give-way control on the minor road, and delineation at the mouth.

9.1 This plan shall be reviewed annually, and following any significant change to the mine's road network.

Appendix B - Intersections requiring positive communication

Nine intersections are listed: the ROM pad turn-off, the RL 360 crossover, the Stage 2 ramp junction, the Stage 2 dump access, the Stage 3 ramp junction, the workshop apron entry, the explosives magazine access, the raw water pump station access and the light vehicle parking bay at RL 400.

Note to Appendix B. This appendix is reviewed and reissued at the annual review of this plan (clause 9.1). An intersection is added to this appendix where it appears on the road layout drawing at Schedule 1 and the review assesses it as requiring positive communication.

The road layout drawing at Schedule 1 is dated 12/11/2025 and is reissued at the annual review. The RL 312 dewatering spur does not appear on it. No provision anywhere in BIN-TMP-004 allows an intersection to be added to Appendix B, or to Schedule 1, between annual reviews, and no provision requires a new road to be referred to the plan's owner when it is cut.

Appendix B is reproduced on a laminated board in each crib room. The board in the Stage 4 crib room is the 12/11/2025 issue.

3. Orrabinda Resources Management of Change procedure ORR-PRO-022, revision 3

5.2 A management of change record shall be raised before any of the following: a change to the mine's road network; the introduction of a new vehicle type; a change to a pit design; a change to a documented control.

5.3 Clause 5.2 does not apply to routine earthworks carried out under a works order within an approved pit design, where no new plant or vehicle type is introduced. Such work is managed under the works order.

5.4 Where a management of change record is raised, the change owner shall complete the impact assessment at Schedule 2, which includes referral to the traffic management plan owner, the survey department for the fleet management system road network, and the contractor's technical services group for the collision awareness system configuration. The three referrals at 5.4 exist only inside the management of change record. The works order form at ORR-PRO-022 Schedule 4 has no equivalent fields, and no referral of any kind.

4. Works order WO-26-4471

Scope: Cut and form 240 m of access road to dewatering bore DW-14 at RL 312, Stage 4 pit, to a formed width of 8 m with windrows to both sides.

Justification: Access required for the commissioning and servicing of DW-14 from 02/08/2026.

Change classification: Routine earthworks within the approved Stage 4 design. No new plant or vehicle type. Management of change not required, ORR-PRO-022 clause 5.3.

Raised by: Mining Engineer, 28/07/2026. **Approved by:** Mining Superintendent, 29/07/2026.

Works completed: 31/07/2026.

Handover: N/A.

The works order carries no scope for signage, no survey pickup of the new intersection, no referral to the traffic management plan, no request to publish a revised fleet management system road network, and no review of the collision awareness system geofences. The handover field is the only place any of them could have been recorded and it reads "N/A".

5. Orrabinda Resources Contractor Management Standard ORR-STD-009, revision 5

7.3 A contractor's own general induction may be accepted as equivalent to the Bindarra general induction where the contractor holds a current recognition agreement with Orrabinda Resources.

7.4 Equivalence under 7.3 does not extend to the Bindarra site-specific supplement, or to the Bindarra light vehicle pit driving assessment, which shall be completed at site by every person who will drive a vehicle in an operating pit and shall not be substituted.

7.5 The engaging department is responsible for verifying that 7.4 has been completed before a site access card is issued.

Hallorn Survey & Geotechnical has held a recognition agreement since 2023. The gatehouse issues site access cards against the recognition agreement alone; it holds no record of the supplement or the pit driving assessment and is not asked for one. Nothing in the access card system distinguishes a person who may drive in an operating pit from a person who may not.

Equipment, roster, training and site records

The investigator's description of what the equipment, roster, training, plan review, hazard reporting and emergency response records show, with wording quoted where it matters.

Fictional source material for the Bindarra case study. The records, the reference numbers, the dates and the figures are invented.

The investigator's description of what the records show, with wording quoted where the wording matters. Records obtained between 13/08/2026 and 26/08/2026 and held in the investigation file.

Haul truck HT-407 maintenance record. Tarowen Mining Services asset register and maintenance history, obtained 13/08/2026. 220 t class rigid haul truck, commissioned 03/2021, 31,480 engine hours. Last 500 hour service 28/07/2026, no defects carried. The monthly brake performance test required by the contractor's standard was carried out on 03/08/2026 and passed, with a measured mean deceleration of 0.80 m/s² net, in the empty condition on a 9 per cent descending grade. The pre-start check sheet for 11/08/2026 is complete and records no defects. A brake performance test carried out on 14/08/2026 after the collision, before the unit returned to service, gave 0.82 m/s² net on the same grade. The retarder, the service brakes and the park brake all met specification before and after the occurrence. No defect was raised against the lighting, the mirrors, the cameras or the cab glass in the twelve months before the occurrence.

Light vehicle LV-118 record. Orrabinda Resources light vehicle register. 4WD utility, in service 02/2024, 61,200 km. Last periodic service 17/07/2026, 4,100 km before the occurrence. Fitted with a certified roll-over protection structure, a 4.5 m amber beacon whip, reflective marking to the site standard, and a UHF radio. The light vehicle inspection sheet for the week of 09/08/2026 records all items serviceable. The beacon lamp was tested after recovery and worked; the whip was sheared at its base in the roll-over.

Haul truck operator, roster and duty record. Tarowen Mining Services roster and site access records. A 14 night on, 7 off roster, working 18:00 to 06:30 local, a 12 hour 30 minute shift with a 30 minute crib. The occurrence was on night 10 of 14, 8 hours 47 minutes into the shift, in its ninth hour, at 02:47 local. Site access records confirm the previous nine shifts worked in sequence with no shift extended beyond 12 hours 40 minutes. The contractor's

fatigue management standard sets a maximum of 14 consecutive night shifts and a minimum 10 hour break, both of which were met. The operator's own sleep record, kept voluntarily and provided by him, records 6 hours 30 minutes of sleep in the day before the occurrence and between 5 and 7 hours on each of the preceding nine days.

Haul truck operator, training and competency record. Fourteen months at Bindarra, seven years on haul trucks across three sites. Verification of competency on the 220 t class unit issued 06/2025 and current. Annual refresher completed 02/2026. Site general induction 06/2025. The record shows no traffic management refresher of any kind since the general induction, and the general induction module on traffic management is a 23 slide package that names the nine Appendix B intersections and does not mention what to do at an intersection that is not listed. There is no training record anywhere on site, for any person, covering the collision awareness system's suppression geofences.

Light vehicle driver, induction and training record. Hallorn Survey & Geotechnical training file and the Bindarra gatehouse access card register. Hallorn general mining induction, a two hour online module, completed 24/07/2026. Bindarra site access card issued 09/08/2026 against Hallorn's recognition agreement, following a 20 minute gatehouse briefing covering muster points, speed limits and personal protective equipment. There is no record of the Bindarra site-specific supplement being issued, completed or signed for. There is no record of the Bindarra light vehicle pit driving assessment. Neither record is required by the gatehouse before a card is issued, and neither was asked for. The driver had worked three night shifts at Bindarra: 09/08, 10/08 and 11/08.

Traffic management plan review history. Document register, BIN-TMP-004. Annual reviews completed 14/11/2023, 10/11/2024 and 12/11/2025, each reissuing Schedule 1 and Appendix B. No out-of-cycle review has ever been carried out under clause 9.1, in the plan's eleven year history. The survey department's road register records 14 new roads cut in the Stage 4 pit since the 12/11/2025 review, of which 11 were cut under a works order and 3 under a management of change record. The three with a management of change record were referred to the plan's owner. None of the 11 was.

Hazard report system extract. Orrabinda Resources reporting system, search of all reports raised at Bindarra in the 24 months to 11/08/2026 containing the terms sight distance, blind, intersection, give way or light vehicle.

Four reports returned, including HAZ-2026-0917. The other three, from 03/2025, 09/2025 and 02/2026, each describe a light vehicle and a haul truck meeting unexpectedly at an intersection not listed at Appendix B, at three different locations. All four were closed by the department that received them, within three days, with an entry recording that the crew had been reminded to take care and that heavy vehicles have right of way. None was referred to the traffic management plan owner. None resulted in a change to Appendix B, to Schedule 1, or to any physical control.

Collision awareness system alert statistics. Tarrowen Mining Services technical services, extract of the fleet CAS event log for the 12 months to 11/08/2026. HT-407 raised 1,684 operator advisory alerts in that period, every one of them outside the S4-LOAD geofence, and none inside it. Across the Stage 4 haul fleet, 68 proximity events were logged inside the S4-LOAD geofence and suppressed. Nine of those 68 involved a light vehicle tag on the main ramp below the geofence boundary, the same mix of vehicles and the same stretch of road as the occurrence; all nine are dated after 09/06/2026, because before the June extension the boundary did not reach the ramp. Before that date the ramp below the crest was outside the geofence and the system was live there; in the ten months of the extract before the extension it raised no alert on the ramp below the crest at night, because there was nothing there to alert on. The suppressed events are recorded in the system and are visible to anyone who queries it. Nobody had queried it: the technical services group's standing report covers alert counts and system health, and does not report suppressed events at all.

Emergency response record. Bindarra emergency response log and the post-incident review, obtained 26/08/2026. The night shift is rostered with one qualified mines rescue paramedic, who was at the surface medical facility. The second responder, required before any extrication, is called in from the accommodation village 11 km from the gatehouse. Dispatch paged the emergency response team at 02:48 local. The first response vehicle entered the pit at 02:51 and reached the scene at 02:54. The second responder arrived on scene at 03:07 in the second response vehicle, which is garaged at the village and carries the hydraulic rescue tool. The driver was extricated through the rear of the vehicle at 03:14, 27 minutes after the collision, and reached the surface medical facility at 03:41. The post-incident review found the extrication was delayed by approximately 9 minutes waiting for the second responder and the tool, measured from 02:58,

when the paramedic's primary survey was complete and the extrication plan made, to the second responder's arrival at 03:07, and recorded that the delay did not affect the clinical outcome on this occasion because the injuries were stable and the patient was conscious throughout.

Interview transcript, haul truck operator

The Teams transcript of the recorded interview with the haul truck operator, with every speaker identified by role.

Lead investigator: We're recording now. Thanks for coming in on your day off. Just for the record, you're happy for this to be recorded?

Truck operator: Yeah, that's fine.

Lead investigator: And you've got your support person here, and you can stop any time you want to. What I'd like first is for you to take me through the shift in your own words. Start wherever you want. I'll hold my questions until you're done.

Truck operator: Alright. It was night ten of fourteen. Pre-start at six, out to the truck about twenty past, and I had 407, same as the night before. It was an ordinary night. Nothing on the pre-start. The dig was on the north side of Stage 4 and we were hauling to the ROM pad, so it's up the ramp loaded, dump, back down empty. We had crib about two. Came back out, did one cycle, and I was on the way back down for the second. Coming down empty you're doing thirty-seven, thirty-eight, you sit it on the retarder and you don't really touch the brakes. That's just how you drive it. There's a crest above the three-twelve. I've come over that crest and there's a glow on the road, and then the whip comes up over it, and it's a light vehicle, coming out of the side onto my road. I got on the brakes and I pulled it right, into the windrow, because that was the only place I had to go. And I hit him. Left front. He went down the hill in front of me and then he went over. I stopped, I called it in, dispatch told me to stay in the cab until they'd stopped the ramp, and then I got out and went down to him. He was talking to me. He was stuck, the door was on the ground. I stayed there with him until the ERT got in.

Lead investigator: Thank you. That's what I needed. Can I take you back to the crest. When you say the whip came up over it, how far away would you say he was?

Truck operator: Not far. I'd say forty metres. It wasn't a long way. It was the sort of distance where you know straight away it's going to happen.

Lead investigator: And what did you do first, in order, as best you remember it?

Truck operator: Brakes and steer. Same movement, really. The retarder was already on, so it was foot brake and wheel. I didn't think about it, I just did it.

Second investigator: Did you consider going left instead, down the open side of the road?

Truck operator: No. If I go left I'm taking him with me, and there's nothing on that side, it's a hundred and something metres down. Right is the wall. Right was the only answer.

Lead investigator: Did you know there was a road joining the ramp at three-twelve?

Truck operator: I knew they'd cut something in there, because you could see the machines doing it a couple of weeks back. I thought it was still closed. There was a closed sign on it.

Lead investigator: When did you last see that sign?

Truck operator: I couldn't tell you a date. It was there, and then at some point it wasn't, and I didn't think anything of it. You're not looking at the side of the road, you're looking at the road.

Lead investigator: What does your in-cab screen show you on that part of the ramp?

Truck operator: The ramp. Just the ramp. There's no road there on the screen. There's no intersection marked, nothing.

Second investigator: Would you expect a new road to appear on it?

Truck operator: I'd expect so, yeah. When they moved the ROM pad tip head last year that was on the screen the next night. That's how you know where you're going.

Lead investigator: Tell me about the collision awareness system. What does it do for you on a normal night?

Truck operator: It sings out when you get near another truck. Up top, on the haul road, you'll get it two or three times a shift, passing someone, coming into the ROM pad. It's a tone and a thing comes up on the screen.

Lead investigator: And down in the pit, on the ramp, below the crest?

Truck operator: Never. I've never had it go off down there. Not once. And I've been on this pit fourteen months.

Second investigator: Did you have a view about why it never goes off down there?

Truck operator: I assumed there was nothing to go off at. You don't see light vehicles on that ramp at night, so if the thing's quiet, the road's empty. That's what I'd have told you the day before. Nobody ever told me it's switched off down there. I found that out from the supervisor, afterwards. That's the bit that's been sitting with me.

Lead investigator: Let's talk about radio calls. What does the site require before you go through an intersection?

Truck operator: There's a list in the crib room. The ROM pad turn-off, the three-sixty crossover, and I think there's seven more on the other pits. You call before you go through those.

Lead investigator: And at an intersection that isn't on the list?

Truck operator: You don't call. Nobody calls. If you called every corner and every turn-off you'd never get off the channel, and dispatch would tell you to get off it.

Lead investigator: Did you make a call before three-twelve that night?

Truck operator: No. It's not an intersection as far as I knew. It's a closed road off the side of the ramp.

Lead investigator: What was the visibility like coming down? Any dust?

Truck operator: There's always a bit of dust. The cart had been through earlier but it dries out. There was some hanging around, yeah. I couldn't tell you if it made a difference. I'm not going to say it did when I don't know.

Second investigator: How were you feeling at that point in the shift?

Truck operator: Tired, the way you are on night ten. Not falling asleep tired. I'd had a decent sleep that day, six and a half hours, which is better than some. I was awake and I was driving properly.

Lead investigator: Is there anything I haven't asked you that you think I should have?

Truck operator: Put a sign on it and put it on the map. That's it. If there'd been a sign on my side of that crest I'd have been at twenty and I'd have been looking for him. And tell people where that system is turned off. I drove that road for fourteen months thinking it was watching for me, and it wasn't.

Lead investigator: That's fair. Thank you. I'll stop the recording there.

Interview notes, night shift supervisor

The investigator's written record of the interview with the night shift supervisor, which was not recorded. These notes are the record of that interview.

Fictional source material for the Bindarra case study. The people, the companies and the records referred to are invented.

Interviewee: Night Shift Supervisor, Stage 4 pit, Tarrowen Mining Services.

Date and place: 19/08/2026, 15:30, by video call from his home during his days off, at a time he chose.

Present: Lead investigator, second investigator. Support person offered and declined.

Recording: None. He was asked and said he would rather talk than be recorded. Notes read back and confirmed, no corrections. These notes are the record of this interview.

Free recall

He has supervised the Stage 4 night shift for Tarrowen for two and a half years, and has 19 years in open pit mining. On night shift he is responsible for the haul fleet, the excavators, the ancillary fleet and the dozers, and for the dispatch operator. He is not responsible for the dewatering crew, the survey contractor, the drill and blast crew when they are on days, or any other light vehicle in the pit.

He was at the ROM pad when the call came over the pit channel. He was on scene about five minutes later. He described the dispatcher's handling as textbook and said he would not change anything about it.

He did not know a light vehicle was in the Stage 4 pit that night. He said this without being asked and returned to it three times during the interview. His words: "Nobody tells me. There's no book, there's no call, there's no permit. If a light vehicle drives past me I know it's there and if it doesn't, I don't."

Additional questions

Did you know the RL 312 spur was open to traffic? Yes. He knew it had been cut and he knew the dewatering crew were using it. He said he had never thought of it as an intersection. "It's a track off the side of the ramp to a bore. There's forty of those in this pit."

Was it ever discussed at a pre-start, on either side? Not that he can

recall, and nothing in his pre-start notes mentions it. He checked his own notes for the ten days between the spur opening on 02/08 and the occurrence and found nothing.

Should the operators have made a call at the spur? He gave a careful answer. His view is that he would want a call. He also said plainly that the plan does not require one there, that he has never pulled anybody up for not calling at an unlisted intersection, and that if he started requiring calls everywhere the channel would be unusable and dispatch would stop him. He said the nine intersections on the board are "the ones we do" and that is the whole of the practice on the pit.

Did you know the collision awareness system was suppressed at RL 312?

Yes, and he was the person who told the truck operator after the collision. He learned about the geofence himself in about the middle of 2025, when he asked the technical services group why the alerts had stopped down at the dig, and was told they had turned them off in that area because the system was going off constantly around the excavator. He said he did not know the geofence had been extended in June 2026 and did not know it now reached the ramp above the spur. He said: "I knew it was off at the dig. I didn't know it was off on the ramp. There's a difference and nobody drew it for me."

****Is there any process at all for a light vehicle entering an operating pit?*** None that he knows of. There is no permit, no radio call-in, no board and no check at the pit entry. Anyone with a site access card can drive in. He raised this himself as the thing he most wants changed.

How was the truck operator in the days after? Not good, in his view, and he has said so to the contractor's health and safety adviser. He asked that this be recorded.

Follow-up questions

- Whether Tarowen's own supervisors have ever been trained on BIN-TMP-004, as distinct from Tarowen's own procedures. He believes they were at some point and could not find a record. Ask Tarowen's training coordinator directly rather than relying on his recollection.
- Whether the technical services group told anybody at Orrabinda, or anybody at Tarowen above him, that the S4-LOAD geofence had been extended in June 2026. He does not know. This is for the technical services group, not for him, and is on the to-do list.

Investigator's note on the interview

The most useful interview of the three for the organisational picture, and the least useful for the sequence, which he did not see. He was frank about the gap between the plan and the practice on his pit, and he did not try to close it with an answer he could not support.

The line worth carrying into the report is his own: he knew the system was off at the dig and did not know it was off on the ramp. The extension in June 2026 carried the boundary up the ramp to a point 180 m above the spur, and nobody outside the group that moved it was told. That is the same pattern as the road itself.

Interview notes, light vehicle driver

The investigator's written record of the interview with the injured driver, which was not recorded at his request and was read back to him at the end. These notes are the record of that interview.

Fictional source material for the Bindarra case study. The people, the companies and the records referred to are invented.

Interviewee: Grade control technician, Hallorn Survey & Geotechnical, driver of LV-118.

Date and place: 21/08/2026, 10:00, at the interviewee's home, nine days after the occurrence and three days after his discharge.

Present: Lead investigator, second investigator, the interviewee's partner as his support person (present, did not participate).

Recording: None. He was asked and declined, and asked instead that the notes be read back to him at the end, which was done. He confirmed them as accurate with two corrections, both made below. These notes, and not a transcript, are the record of this interview.

Fitness: He was in a sling and tired easily. The interview ran 55 minutes with two breaks at his request.

Free recall

He has been a grade control technician for six years, the last four with Hallorn. He works a 9-on 5-off roster and rotates between four client sites. Bindarra was a new site for him. The occurrence was on the third night shift of his first swing there.

His task that night was a pickup at the new dewatering bore collar at RL 312: a set of levels on the collar and the pad, for the dewatering department's as-built. It was his third night working at that bore. He had driven the spur six or seven times over the three nights, at various times of night, and had never met a haul truck at the intersection or anywhere on the ramp below the crest.

He left the bore collar at about ten to three, drove the 240 m out to the ramp, slowed at the mouth of the spur, looked right up the ramp, saw nothing, and drove out, turning right to go back up to the surface. He does not remember hearing anything on the radio. The next thing he remembers is light on his right and the noise, and then being on his side with the door under him. He remembers the truck operator talking to him through the windscreen and telling him not to move.

[Correction on read-back: he first said "stopped at the mouth" and then

corrected it himself to "slowed", and asked that the note say he does not believe he came to a complete stop.]

Additional questions

What could you see up the ramp when you looked? Not much. The windrow is along the side of the spur and the ramp goes up over a rise, so you can see a piece of road and then it's gone. He said he had assumed that if a truck were coming he would see the lights on the wall first, and that this had worked the other times.

Was there a give-way sign, a stop line, or any marking at the mouth?

No. He was clear on this and said it twice. There was a road closed sign when the road was being built and it was gone by the time he started using it.

How fast were you going as you came out? "Twenty-five, maybe. Not fast. It's a ramp, you don't go fast." He agreed he did not stop.

What were you told about radio calls in the pit? Nothing specific. He had a radio in the vehicle, it was on channel 12 because that is what the dewatering foreman told him to put it on, and he had it turned up. Nobody told him to make a call before entering the ramp, and he had never heard anybody else make one. He said if he had been told to call, he would have called. [Correction on read-back: he asked that "I would have called" be recorded in his own words rather than paraphrased.]

What induction did you do for Bindarra? A two-hour online module at the Hallorn office in town, which he described as "the general mining one", and a site access card issued at the Bindarra gatehouse on his first day after a twenty minute talk about the muster points and the speed limits. He did not do the Bindarra light vehicle pit driving assessment and had not heard of it. He has never seen the Bindarra traffic management plan and did not know there was one.

Had anyone told you the spur was open to light vehicles? Yes, the dewatering foreman, on the first night, who drove him out there himself.

What was the visibility like? He was firm about this. The air was clear, he could see the ramp surface and the windrow fine, and there was no dust hanging in the spur. He said the dust at Bindarra is "nothing like" the other sites he works at.

Anything about the vehicle? Nothing. It drove normally, the lights

worked, the beacon was on because he switches it on at the gatehouse and never switches it off. The seatbelt was on.

Follow-up questions

Still open with this interviewee, for a second conversation once he is further along:

- Whether he was ever issued, or asked to sign for, the Bindarra site supplement to the general induction. He could not remember and did not want to guess. Chase Hallorn's training records first and then put the answer to him rather than asking him to recall it.
- Whether anybody at Bindarra, at any point, asked him what work he was doing in the pit that night or where he would be. He began to answer and became distressed. Not pressed.

Investigator's note on the interview

He is a careful witness. He corrected himself twice without being pushed, including on the point that is least helpful to him, and he declined to guess on two questions where a guess would have been easy. His account of the sight line matches the survey to within a few metres, which is a good check on the rest of it.

His evidence on the dust directly contradicts the truck operator's. Both accounts are honest and neither of them is a measurement. He was stationary or slow in the spur; the truck operator was descending a ramp behind his own lights. They may both be right about what they could see, which is why the dust question cannot be settled on what either of them said.